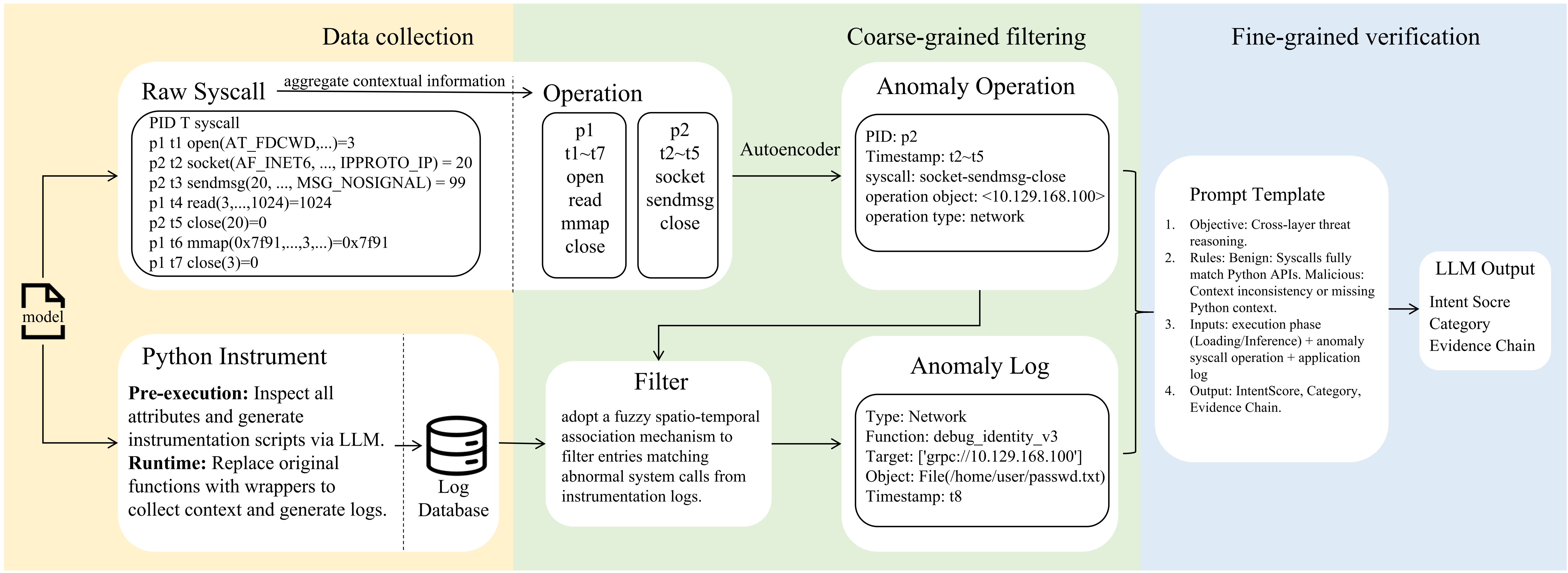
Data collection
Coarse-grained filtering
Fine-grained verification
Raw Syscall
aggregate contextual information
Operation
PID T syscall
p1 t1 open(AT_FDCWD,...)=3
p2 t2 socket(AF_INET6, ..., IPPROTO_IP) = 20
p2 t3 sendmsg(20, ..., MSG_NOSIGNAL) = 99
p1 t4 read(3,...,1024)=1024
p2 t5 close(20)=0
p1 t6 mmap(0x7f91,...,3,...)=0x7f91
p1 t7 close(3)=0
p1
t1~t7
open
read
mmap
close
p2
t2~t5
socket
sendmsg
close
Autoencoder
Anomaly Operation
PID: p2
Timestamp: t2~t5
syscall: socket-sendmsg-close
operation object: <10.129.168.100>
operation type: network
model
Python Instrument
Pre-execution: Inspect all attributes and generate instrumentation scripts via LLM.
Runtime: Replace original functions with wrappers to collect context and generate logs.
Log Database
Filter
adopt a fuzzy spatio-temporal association mechanism to filter entries matching abnormal system calls from instrumentation logs.
Anomaly Log
Type: Network
Function: debug_identity_v3
Target: ['grpc://10.129.168.100']
Object: File(/home/user/passwd.txt)
Timestamp: t8
Prompt Template
1. Objective: Cross-layer threat reasoning.
2. Rules: Benign: Syscalls fully match Python APIs. Malicious: Context inconsistency or missing Python context.
3. Inputs: execution phase (Loading/Inference) + anomaly syscall operation + application log
4. Output: IntentScore, Category, Evidence Chain.
LLM Output
Intent Socre
Category
Evidence Chain